

LED TV Head with Expressions



Version 1.0

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Intro

Whether you're going wear it at a convention, around town, or just have a cool room decoration, this build provides the code to add expressive LED eyes to whatever you'd like!

This project is built with an Arduino Nano and uses two WS2812B 8x8 LED displays for the eyes. The expressions are controlled by a Nintendo Wii Nunchuk which can cycle through 16 different expressions (plus 1 idle expression).

This guide also provides steps on how to import your own custom expressions!

Whether you have no experience with DIY projects or are a hobby veteran, I hope this guide proves to be helpful. Happy making!

Materials

This project is based around the Arduino Nano, an inexpensive and readily available microcontroller. This will be the heart of the system, holding all of the code and expressions.

- Arduino Nano w/ATMega328p
 - Because the program is just over 16kb, you **must** use a microcontroller such as the ATMega328p (32kb of storage) that has more than 16kb of storage
 - I recommend using a USB-C variant for convenience. Though, micro-USB and mini-USB Arduinos will work the same
- WS2812B 8x8 Matrix Display x 2
 - Awesome little RGB displays!
 - You will need two of them for both eyes
- Nintendo Wii Nunchuk
 - A convenient and cheap controller for Arduino projects
 - Can use first-party or third-party nunchuks
- Portable Rechargeable Battery
 - Any cheap rechargeable battery pack will work fine

Specs for Nerds

The code is built using the FastLED library written by the FastLED team and the [Nunchuk.h](#) library written by infusion/Robert Eisele.

FastLED is a fast and efficient library used for controlling a wide variety of different LED chipsets. Here, it is set to use the WS2812B chipset.

The controller inputs are handled by the Nunchuk.h. Again, a quick and easy to use library which even provides the nunchuk's accelerometer data (perhaps for more ways to change the expressions?).

The expressions are stored in the PROGMEM of the Arduino.

Making the Electronics

Plug in the Arduino to your PC and upload the code. Don't know how to upload the code? Follow [this tutorial](#) or follow these steps:

Step 1. Download Arduino IDE from

<https://www.arduino.cc/en/software>. Download and install the latest version on your computer.

Step 2. Setup

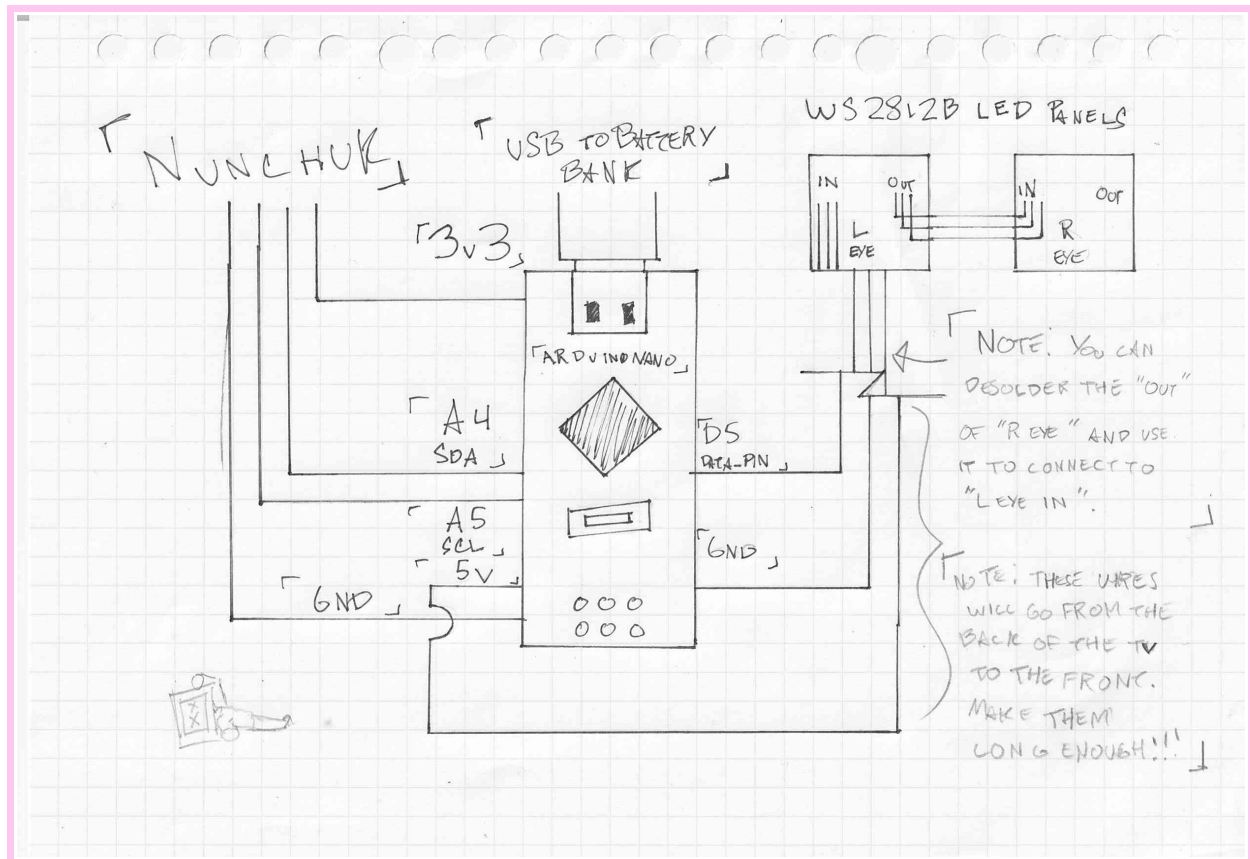
Open the software and plugin the Arduino to a USB port. Under the “Tools” tab, navigate to *Board: > Arduino AVR Boards > Arduino Nano* (or whatever Arduino you are using). In the same menu, click “Processor:” and select the variant with “(Old Bootloader)”. Lastly, click “Port” in the same tab and select whichever your Arduino is connected to.

Part 3. Open and Upload

Click “File” in the top left corner and open the included file “tvhead_nuncntrl.ino”. Click the right-facing arrow in the top left to upload your code and wait for it to finish. Your Arduino is now ready to go!

Once done, follow this wiring diagram and get soldering! You can solder directly to the board or use a circuit board and 3D printed case to keep things more organized.

(NOTE: Please follow the [Nunchuk articles guide](#) for the coloring of different wires. Not all Nunchuks have the same wire colors!)



Editing and Adding Faces

There are many parts you can tweak and modify in the code. Here's a few major ones which might be handy if you're looking to customize or modify the code:

Brightness

```
18 // BRIGHTNESS, from 0 to 255, lower number = less power usage = longer battery life!  
19 #define BRIGHTNESS 32
```

You can easily change the brightness here. Brighter LEDs may look nicer but also will use up more power.

Joystick Threshold

```
30 // threshold, from 0 to 127, lower means more sensitive  
31 // Default value = 120  
32 const int threshold = 120;
```

You can change how sensitive the joystick is to change expressions. Lower number == more sensitive. Higher number == less sensitive.

Expressions

Expressions can be made using the included tool. Open it in any spreadsheet editor and draw according to the diagram. Use the "Idx" section as your color palette and draw by using the corresponding index value. You have to draw things sideways for them to appear normal.

Once you've finished drawing, copy the “Data Table 8 x 16 (TV Head)” section into the Arduino code images section. Be sure to add the code surrounding it and replace “NAME” with the name of your new expression:

```
const long NAME[] PROGMEM = {

// PASTE YOUR IMAGE DATA HERE

};
```

Then scroll down to the switch...case near the end of the program and find the control you want to assign your new expression to (ex. Idle or C button + Up). Again, replace “NAME” with the name of your expression:

```
579 | case 4:
580 |     // down
581 |     for (int i = 0; i < NUM_LEDS; i++) {
582 |         leds[i] = pgm_read_dword(&NAME[i]);
583 |     }
584 |     FastLED.show();
585 |     break;
```

Huzzah! You've now added your own expression. Don't forget to upload the code to the Arduino.

Credits and Links

nunchuk.h Library by infusion/Robert Eisele

<https://www.xarg.org/2016/12/using-a-wii-nunchuk-with-arduino/>

LED Matrix Excel Spreadsheet by marco_c

<https://arduinoplusplus.wordpress.com/2019/11/16/easy-neopixel-bitmaps-using-excel/>

FastLED by The FastLED Team

<https://github.com/FastLED/FastLED>